IND5003-2610 Project Brief

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1 Project Titles

1.1 Airline Delay

Goal

The goal of this project is to study flight delay propagation in US domestic flights.

Data

The Bureau of Transportation Statistics (BTS) provides flight operation details for every flight, going back to 1987. A sample of the data (for two months) can be found in our course teams/sharepoint folder. The columns in the data include:

- Scheduled departure and arrival
- Actual departure and arrival
- Flight details, e.g.:
  - Tail number of plane
  - Airline
  - Origin and destination airport codes
  - Cause of delay, if any

Take note, after unzipping, the data for each month occupies approximately 250MB of disk space.

Questions of Interest:

1. Suppose we represent the network of flights using a graph, with airports as nodes and flights as edges. Which airports act as delay amplifiers/absorbers?
2. Once we know a flight has been delayed at a particular airport, can we predict the downstream delays?

3. The dataset assigns the cause of a delay (air carrier, weather, NAS, late-arriving aircraft, and security). Given the delay cause and delay time, is it possible to attribute the delay to upstream airports / flights?

Challenges & Expectations:

1. The two papers provide an overview of techniques used in this very operational problem. Apart from the domain, depending on the analysis you choose, you may need to learn new techniques, such as network analysis or epidemic modeling (simulation).
2. You will have to decide for yourself how many months of data to include in your study. Handling the huge dataset will be a challenge. Apart from this, you will have to look for tables with information on the planes, airports and weather.
3. Feature engineering, e.g. what indices to use to measure various aspects of a delay.
4. When training an ML model, be sure to use time sensitive train/test splits, like in time series models. Random splits may lead to data leakage!

References:

1. It is very important to read at least these articles to understand the possible approaches you can take:
   - Flight delay propagation modeling: Data, Methods, and Future Opportunities
   - Network science approach for identifying disruptive elements of an airline
   - Air Travel Delay Prediction: Feature Engineering and ML Approaches (UC Berkeley)
2. The following pages on the BTS will be useful as well:
   - Data download
   - Understanding cause reporting
   - Air carrier statistics

1.2 Time Series

Goal

The goal of this project is to forecast the monthly number of tourist arrivals into Singapore for the calendar year 2027.

Data

The Singapore Tourism Board (STB) and data.gov.sg both provide monthly arrivals from all countries (by sea, land and air), dating back to Jan 2008. Other datasets, that you will need to look for on your own, include:

- Public holiday calendars from visiting countries.
- Hotel statistics, such as expenditure.
- One-time event dates, such as:
  - Concerts
- Regular events such as Formula One race.

Questions of Interest

1. How accurately can we forecast total monthly arrivals into Singapore? Which countries arrival patterns are similar to one another?
2. Can we quantify the impact of public holidays in the region, including long weekends?
3. Is it possible to forecast the effect of one-off events such as:
   - Taylor Swift concert, which should lead to a bump in arrivals.
   - Pandemic, which could lead to a prolonged depression in arrivals.

Challenges & Expectations

1. As part of this project, you will need to learn a few time series techniques on top of the ones in the textbook. These may include:
   - X-13ARIMA-SEATS model, which adjusts for holiday variation and external regressors.
   - Hierarchical time series modeling, which cater to the forecasting of a collection of time series that aggregate in some way.
   - You could also explore deep/shallow learning models for forecasting.
2. Take note that although the end goal of the project is to forecast arrivals for the year 2027, you will have to back-test your model on held out data, e.g. on 2025 and 2026, and report the performance.
3. Remember to pay close attention to explainability of your model, and to reproducibility.
4. If it is possible to tie tourist arrivals to economic boost in some way, that would make the analysis complete.

References

1. X-13ARIMA-SEATS model
2. Forecasting hierarchical time series

1.3 Natural Language Processing

Goal

The goal of this project is to analyse Annual General Meeting (AGM) minutes from Singapore Exchange (SGX) listed companies within a single sector. The recommended sector to work with is the telecommunication industry, but you can switch to one of the others if you are more familiar with it:

- Banking
- Property developers
- F&B

Data

Data for this project can be downloaded from the investor relations sections of the relevant listed companies. Most of the reports are in pdf format.

Historical stock prices can be obtained directly from Yahoo finance.

Questions of Interest

1. How does the meeting content differ across companies and across time, within the same sector?
2. Does the content affect the stock price (or vice versa) of a company, or others in the sector?

Challenges & Expectations

1. The team will be expected to perform a natural language analysis of the meeting minutes across companies and across time. Include at least the following aspects:
   - Named entities mentioned
   - Sentiment/emotion (of shareholder questions, management responses, overall tone)
   - Topics/themes discussed
2. Correlate the extracted signals with the stock performance.
3. For more recent years, the text extraction converting the AGM minutes into text can be performed easily using Python packages. For older years, meeting minutes may have be scanned, resulting in images.
   - This may require OCR on your own, or you may elect to use an off-the-shelf visual LLM (using an openrouter key from us).
   - If you use the latter, ensure that you have a section in your report that validates the transcription from the LLM.
4. A large part of AGM minutes consists of standard procedural text. Your pipeline should isolate this before analysis.

5. There are numerous huggingface NLP models, and you may also elect to use openrouter AI models. But it is imperative that you validate the classification/tagging/sentiment results, since the models were not trained on AGM data.
6. For this project, the major emphasis should be on the interpretation, and correlation with stock prices/analyst sentiment etc. The following steps are necessary but routine; they will account for the least credit:
   - Assessing or improving the accuracy of the transcription,
   - Fine-tuning the accuracy of an embedding, or of the sentiment classification.

References

1. Example of AGM minutes website

1.4 Vision I

Goal

The goal of this project is to study the build-up of traffic jams at the Singapore-Johor land connections (Woodlands Causeway and Tuas Second Link)

Data

The course sharepoint folder contains sample images, but you *must* collect more images during the course of the semester. Take note that LTA only stores the images for five minutes, so you will have to set up a script to run every five minutes.

Questions of Interest

1. How early can we detect a traffic jam build up at the two links?
2. Once there is a traffic jam, how accurately can we predict the length of time the traffic jam will continue for, and when it will dissipate?

Challenges and Expectations

1. Image processing is an important part of this project. You are expected to attempt several approaches to improve the accuracy of vehicle count detection from the images. Training a new detection model may be necessary, but is not the only focus.
2. The identification of traffic jams should be based on a 5-minute time series of counts, but other regressors or information could be incorporated, e.g. weekend, public holiday, time of day and weather information.
3. Your model and technique should be tested on new data, collected this semester, using the code in the shared notebook. If necessary, register for your own LTA API key for free.
4. As ground truth indications of traffic jams, you can use the Traffic Speed Bands, Estimated Travel times (both from LTA API), or ICA advisories.
5. *Do not share data across groups*; plan to collect your own.
6. Ensure that the teaching team can reproduce your results easily.

1.5 Vision II

Goal

This project requires you to build a pipeline for feature extraction/engineering.

Data

The data consist of screen recordings of penalty shootouts at the FIFA 2018 world cup. They can be downloaded from our course sharepoint folder.

Questions of Interest

The initial portion of this project requires you to develop scripts that pre-process the videos.

1. Given a penalty event (kicker walks to spot until play ends), the script should aim to extract the following features:
   - Related to kicker:
     - Trajectory of kicker's run-up (top-down view)
     - Right/left foot kick
     - Number of steps taken
   - Related to goalkeeper
     - Movement before kick
     - Dive direction
     - Dive height
   - Related to ball
     - Ball trajectory
     - Final position
   - Related to outcome
     - Goal/no goal
2. Similarly, given a penalty event, a separate script should return the timings for the main camera angles:
   - From back of kicker
   - From 45 degree angle (kicker's right)
   - Slow motion close up of kicker
   - Slow motion from back of goalkeeper

Challenges & Expectations:

1. Since there are only four penalty shootouts, data is scarce. The onus is on the team to study the footage to develop and identify features.
2. Video slicing can be done manually, e.g. to split the sequence into separate penalties. However, everything else must be done through a script, i.e. must be reproducible.
3. Upon extraction of features, your team is expected to perform a basic EDA of the penalty kicks.

References

3. The following github repository contains information on existing benchmarks for the accuracy of feature extraction from broadcast soccer matches.

2 Submissions

The following are the mandatory submissions for the group project:

1. Video presentation
   - A 10-minute video to be uploaded to a sharepoint folder (to be created closer to the submission date).
   - The presentation should summarise the work done, and the major findings from your work.
   - You can assume that the audience for the presentation is a technical audience, i.e. it is a knowledge-sharing presentation.
2. Report
   - A detailed report of the work done, submitted as a pdf document.
   - The maximum number of pages should be 25, excluding cover, contents, references and appendices. The font size cannot be smaller than 10 Times New Roman.
   - See below for more details.
3. Source codes
   - All codes and data should be uploaded to sharepoint.
   - Codes can be in the form of python notebooks, or in the form of scripts/modules, or a combination of both.

The deadline for the above submissions is *30th Oct 2359hrs*.

3 Grading

The total marks for the project is 55% of your final grade. This is broken down into 40% for a common group grade, and 15% for an individual grade.

3.1 Group grade (40%)

Video presentation (10%)

The video will be graded based on clarity of content, use of visuals, and creativity. The allowed time for the video is quite short, so you will have to plan the presentation and content properly. Here are some tips regarding the content of the video:

- Provide a summary of the findings upfront, and then go into the details.
- Use graphs and images to explain your points, rather than text-heavy slides.
- Do not speed up your video. You will be heavily penalised if this is suspected.
- Not everyone needs to narrate the video. You can choose just one person. However, avoid using an AI voice.
- If you choose to include your face/headshot in the video, do not read from a script. Speak naturally; it is ok to make a mistake here and there.

Project report (20%)

Your report *must* include the following clearly demarcated sections:

1. Literature review of similar approaches.
2. Data overview. Reproducibility is an integral component of your project. This section is where you describe the data collection and preparation procedures. Remember that a good data analyst will take the initiative to look for additional data to strengthen the findings. Credit will be given for finding relevant datasets that provide additional evidence/context for your conclusions.
3. Exploratory data analysis. This section should guide the direction of your subsequent analysis.
4. Statement of questions of interest. These are the 2 - 3 research questions that you aim to answer with this analysis.
5. Overview of new techniques. If you learnt new techniques (defined to be methods that are not present in the textbook), include a section where you provide an overview
6. Overview of code structure

The data analysis report will be graded on the quality of the analysis and the clarity of the report. In terms of clarity of report, we look for the use of proper English (no bullet points). We also encourage the use of visualisations.

The report should be well-organised. In terms of quality of analysis, we look for *iteration* in the analysis, since every analysis reveals a bit more about the data, and suggests what could be done next.

The findings pertaining to the 3 questions of interest should be communicated clearly, demonstrating understanding of techniques. A discussion of the limitations of the analysis should also be present.

Evidence of self-learning, where you identify a new technique, or go deeper into one of the techniques introduced in the lectures) also counts as a plus point. A very insightful paper by prominent statisticians (McGowan et al. (2023)), breaks down every data analysis into six areas that can be used to assess its quality. These six areas are: data quality, exhaustiveness, skepticism, second-order questions, clarity and reproducibilty of analysis. You may want to skim through the paper when assessing the quality of your own report.

Please note that NUS has a policy on the use of AI for homework. I encourage you to use AI models to assist with writing code, but do ensure that you do not violate the university guidelines. Especially when writing reports, bear in mind that these models do tend to invent references, authors and titles.

Remember that you are responsible for every line of code, every sentence in your report and every statement in your video.

Code submissions (10%)

Code is assessed in terms of reproducibility, documentation (via comments/docstrings/README files), and organisation of folders.

Remember that reproducibility is an integral part of this project. The teaching team must be able to reproduce your results and run your code *easily*. Toward that end, please do not use personal accounts (e.g. personal AWS, free cloud credits, AI accounts, database platforms) in your submission. Please see below for the cloud computing options we have for our course.

3.2 Individual grade (15%)

There are two components to the individual grade:

1. An individualised Quiz, based on the work that your group has submitted. (10%)
2. A peer evaluation of your individual contributions in the project. This will be administered using a survey via the TEAMMATES application. After the project submissions, you will be required to answer 2 - 3 questions about your teammates, and to then summarise their individual contributions. (5%)

   If you do not submit this survey on time, you will receive 0% for this component even if you have contributed in the project.

4 Available computing/AI resources

If you feel you need more dedicated computing resources for your solution, let us know. Our department has a license with Ronin platform. This means that you can easily start AWS EC2 instances (for compute), and use S3 services (for storage), and keep to a budget.

If you need AI models in your solution, e.g. you need to call an Anthropic AI model to perform summary of a document, or OCR extraction, then we have a budget for using OpenRouter keys. OpenRouter API keys allow you to call OpenAI/Anthropic/Google/many other AI models through Python. If you need this approach, we will issue you an OpenRouter API key with a budget for your use this semester.

Please contact the teaching team if you need more information about the above.

References

McGowan, Lucy D'Agostino, Roger D Peng, and Stephanie C Hicks. 2023. "Design Principles for Data Analysis." *Journal of Computational and Graphical Statistics* 32 (2): 754–61.